

Universitat Politècnica de València
Master in Artificial Intelligence, Pattern Recognition and Digital Imaging
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MACHINE TRANSLATION

5. Advanced Topics in Neural Machine Translation

Francisco Casacuberta
`fcn@prhlt.upv.es`

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Word embeddings

- Character, sub-word, word and sentence embeddings: vector representation.
- Many NLP applications.
- Word embeddings from monolingual corpus: i.e. word2vec (Sec. 2, Chap. 3).
- Word embeddings from a training process: Sub-product of sequence-to-sequence training (Sec. 4-6, 3).
- Word embeddings from a character embedding using CNN (Sec. 7, Chap. 3).
- Word embeddings from pre-trained LMs: i.e. BERT (topic 5 in Chap. 5) or ELMo [Peters NAACL 2018]
- Multilingual word embeddings: Represent words from multiple languages in a single distributional vector space [Chen+ EMNLP 2018].

A common vector space for word embeddings

[Conneau+ ICLR 2018][Artetxe+ ACL 2018]

- Learn a linear mapping $\widehat{W}$ between the source W_X^E and the target W_Y^E embeddings:

$$\widehat{W} = \underset{W}{\operatorname{argmin}} \|W W_X^E - W_Y^E\|$$

- Align monolingual word embeddings.
 - Supervised, from a train bilingual dictionary.
 - Unsupervised using adversarial training,
- Toolkit **MUSE** [Conneau+ ICLR 2018]
<https://github.com/facebookresearch/MUSE>
- Toolkit **VecMap** [Artetxe+ ACL 2018]
<https://github.com/artetxem/vecmap>

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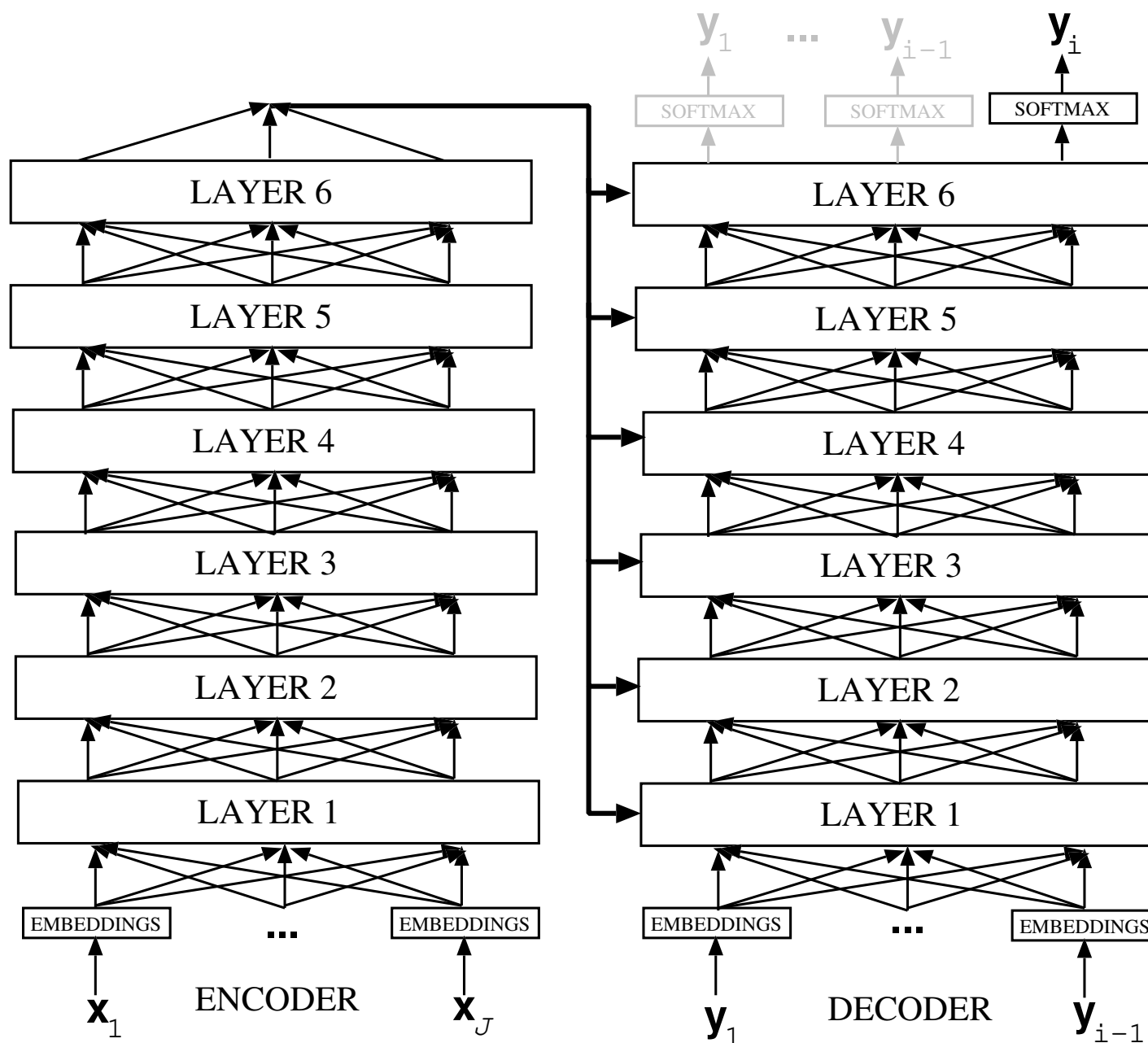
Why pre-trained models?

- Training neural machine translation models from scratch is expensive.
- Large bilingual corpora are necessary.
- For many task-specific translation models there are low resources.
- Pre-trained models have proved to be useful in many scenarios.

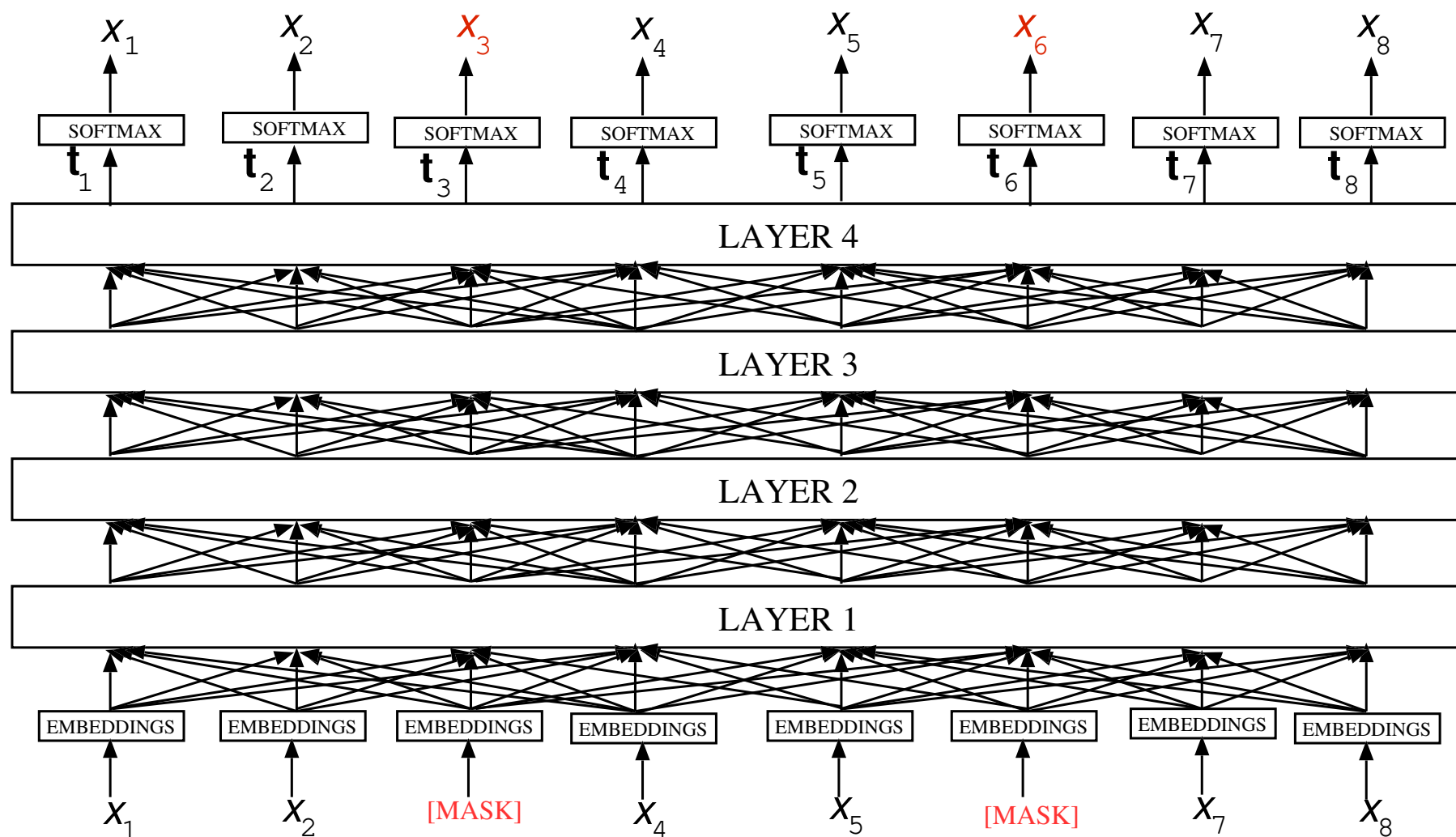
Pre-trained models

- Bidirectional Encoder Representations from Transformers (BERT). Based on the Transformer encoder. [Devlin NAACL 2019] -Google AI Language-
- Generative Pre-trained Transformer (GPT, GPT-2 and GPT-3). Based on the Transformer decoder. [Radford NAACL 2018] -OpenAI-
- Generalized Autoregressive Pretraining (XLNet). Decoder of Transformer without masks + permutation language modelling. [Yang NIPS 2020] -Google AI Brain-
- Cross-lingual Language Model Pretraining (XLM). Transformer-based [Conneau NIPS 2019] -Facebook AI-
- Random Aligned Substitution Pre-training (mRASP). [Lin arXiv 2021]
- A survey: [Sun SCTS 2020]

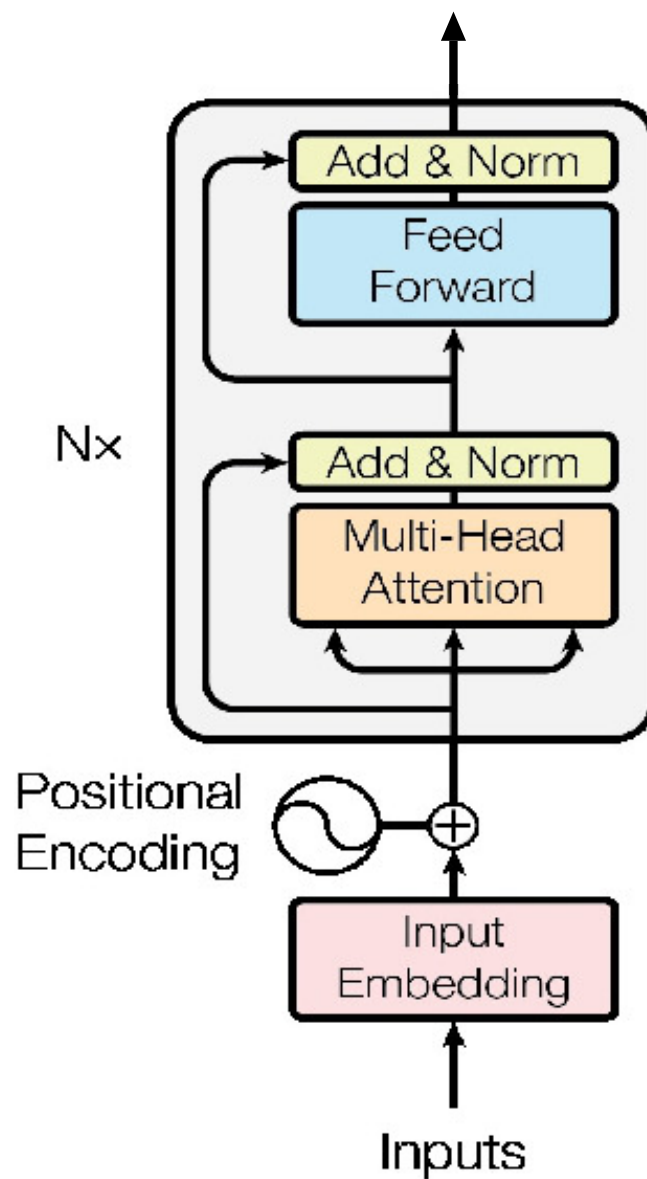
Transformer model



BERT model



Encoder in the Transformer (Vaswani 2017)



The layers in BERT (encoder of Transformer)

Given a source sentence x_1^J ,

- Initialization: $\mathbf{h}_j^0 = \mathbf{x}_j = \mathcal{E}(x_j) \quad 1 \leq j \leq J$
- In layer l of the encoder ($1 \leq l < L$)
 - Self-attention model:

$$\mathbf{u}_j^{l+1} = \mathbf{N}(\mathbf{R}(\mathbf{a}(\mathbf{h}_1^l, \dots, \mathbf{h}_J^l, \mathbf{h}_j^l), \mathbf{h}_j^l)) \quad 1 \leq i \leq J$$

- Feed-forward network:

$$\mathbf{h}_j^{l+1} = \mathbf{F}_f(\mathbf{u}_j^{l+1}) \quad 1 \leq j \leq J$$

The input of BERT [Devlin NAACL 2019]

Word embeddings: For each input token sum the token embedding plus segment embedding (for two input sentences) plus position embedding.

Embeddings	$\mathbf{x}_0$	$\mathbf{x}_1$	$\mathbf{x}_2$	$\mathbf{x}_3$	$\mathbf{x}_4$	$\mathbf{x}_5$	$\mathbf{x}_6$	$\mathbf{x}_7$	$\mathbf{x}_8$	$\mathbf{x}_9$
Positional embedding	E_0^p	E_1^p	E_2^p	E_3^p	E_4^p	E_5^p	E_6^p	E_7^p	E_8^p	E_9^p
	+	+	+	+	+	+	+	+	+	+
Sentence embeddings	E_0^s	E_1^s	E_2^s	E_3^s	E_4^s	E_5^s	E_6^s	E_7^s	E_8^s	E_9^s
	+	+	+	+	+	+	+	+	+	+
Tokens embeddings	$E_{[\text{CLS}]}^t$	$E_{x_1}^t$	$E_{x_2}^t$	$E_{[\text{MASK}]}^t$	$E_{x_4}^t$	$E_{[\text{SEP}]}^t$	$E_{x'_1}^t$	$E_{[\text{MASK}]}^t$	$E_{x'_3}^t$	$E_{[\text{MASK}]}^t$
Tokens	[CLS]	x_1	x_2	[MASK]	x_4	[SEP]	x'_1	[MASK]	x'_3	[MASK]
Input		x_1	x_2	x_3	x_4		x'_1	x'_2	x'_3	x'_4
	————— sentence A —————						————— sentence B —————			

$$\mathbf{x}_j = \mathcal{E}(x_j) = E_{\bar{x}_j}^t + E_j^p + E_j^s \quad 0 \leq j \leq 9$$

$$E_j^s = \begin{cases} E_A & 0 \leq j \leq 5 \\ E_B & 6 \leq j \leq 9 \end{cases}$$

Masked language model [Devlin NAACL 2019]

- Training masked language with BERT:
 - Masked language model: during training mask some percentage of the input tokens at random, and then predict those masked tokens.
 - In 15% of positions at random:
 - * 80% the token is substituted by [MASK].
 - * 10% the token is substituted by another random token.
 - * 10% the token remains unchanged.
 - The output corresponding to [CLS] is used for classification if the second sentence follows the first one (by adding a layer: a feed-forward classifier)
 - Slow convergence.
- Fine tuning with BERT:
 - For classification: adding a classifier layer in the correspondent output of the [CLS] token

Masked language model

- Given a sentence x_1^J , with $x_j \in \mathcal{V}_X$ for $1 \leq j \leq J$, let $\mathcal{J} \subset \{1, \dots, J\}$.
- A masked sentence of x_1^J by $\mathcal{J}$ is a sentence $\bar{x}_1^J$, such that:

$$\bar{x}_j = \begin{cases} [\text{MASKED}] & j \in \mathcal{J} \\ x_j & \text{otherwise} \end{cases} \quad \text{for } 1 \leq j \leq J$$

- The probability of the masked words is:

$$p(\{x_j : j \in \mathcal{J}\} \mid \bar{x}_1^J) = \prod_{j \in \mathcal{J}} p(x_j \mid \bar{x}_1^J) = \prod_{j \in \mathcal{J}} \mathbf{f}_{sm}(\mathbf{W} \mathbf{h}_j^L)_{i(x_j)}$$

BERT model [Devlin NAACL 2019]

- # Layers=12 or 24; # layer size = 768 or 1024; # self-attention heads = 12 or 16.
- Sentence embedding:
 - The first output token corresponding to [CLS] previous to the softmax operation.
 - The sum of the word embeddings $\sum_{j=1}^J \mathbf{t}_j$
 - The mean the word embeddings $\frac{1}{J} \sum_{j=1}^J \mathbf{t}_j$
- BERT as a Markov Random Field Language Model [Wang NeuralGen 2019]
- Extension: Multilingual BERT (M-BERT) [Pires ACL 2019]
 - Monolingual text of Wikipedia from 104 languages.
 - Shared Word-piece vocabulary.
 - Good for zero-shot cross-lingua model transfer.
 - No marker of language is used.
 - In [Chi ACL 2020] “Subspaces of mBERT representations recover syntactic tree distances in languages other than English, and that these subspaces are approximately shared across languages”.

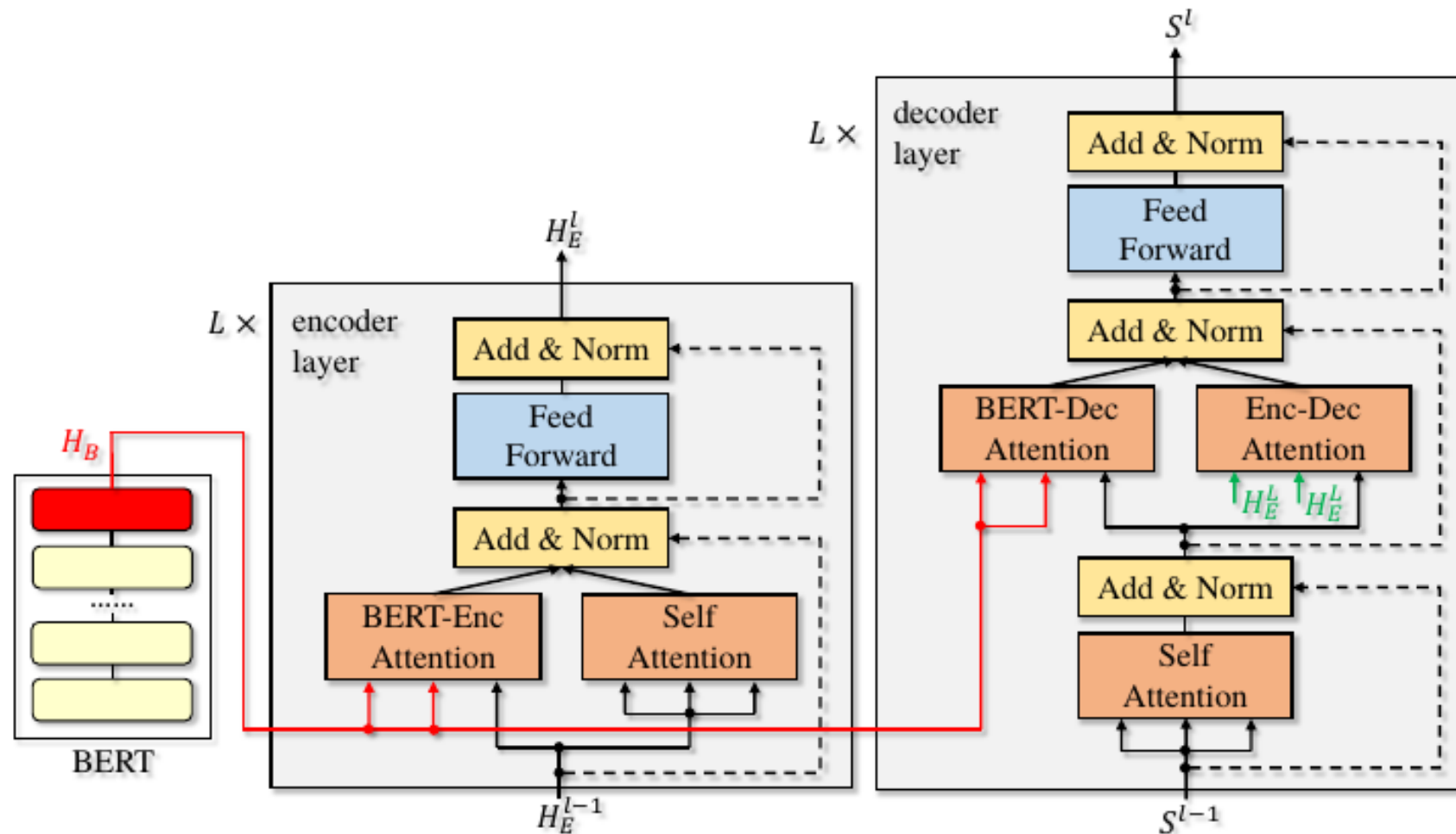
Other pre-trained BERT-like models

- ALBERT: A Lite BERT [Lan arXiv 2019]
- SBERT: Siamese and triplet network structures [Reimers EMNLP 2019]
- BART: A denoising auto-encoder based on an encoder-decoder architecture [Lewis ACL 2020]
- RoBERTa: A Robustly Optimized BERT Pretraining Approach. [Liu arXiv 2019]
- Char-Bert; BERT at character level [Ma COLING 2020]
- DistilBERT: a smaller general-purpose language representation model [Sahn NeurolPS 2019]
- ExpBERT, GAN-BERT, MobileBERT, DeeBERT, schuBERT, SentiBERT, BERTRAM, CluBERT, MTSI-BERT, SiBERT, FlauBERT, NegBERT, AraBERT, BioBERT, SciBERT, ClinicalBERT, TransBERT, DocBERT, PatentBERT, XLNet, SpanBERT, ... [ACL 2020][LREC 2020]

Other applications of pre-trained models

- Pretrained models for MT:
 - BERT-fuse model [Zhu ICLR 2020] [Yang AAAI 2020]
 - Cross-lingual language model (XLM) [Conneau NIPS 2019]
 - Different initializations of Transformer with BERT [Clinchant WNGT 2019].
- Multilingual machine translation.
- Document summarizing.
- Text classification.
- Visual question answering.
- Image and video description.
- Next sentence prediction.

BERT-fused for machine translation [Zhu ICLR 2020]



BLEU in IWSLT'14 En→De

Standard Transformer	28.6
Bert-fused model	30.5

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Why multilingual translation?

- For low-resource languages.
- For no resources: zero-shot translation models.
- To take profit of common features in similar languages.
- A single engine instead many engines.

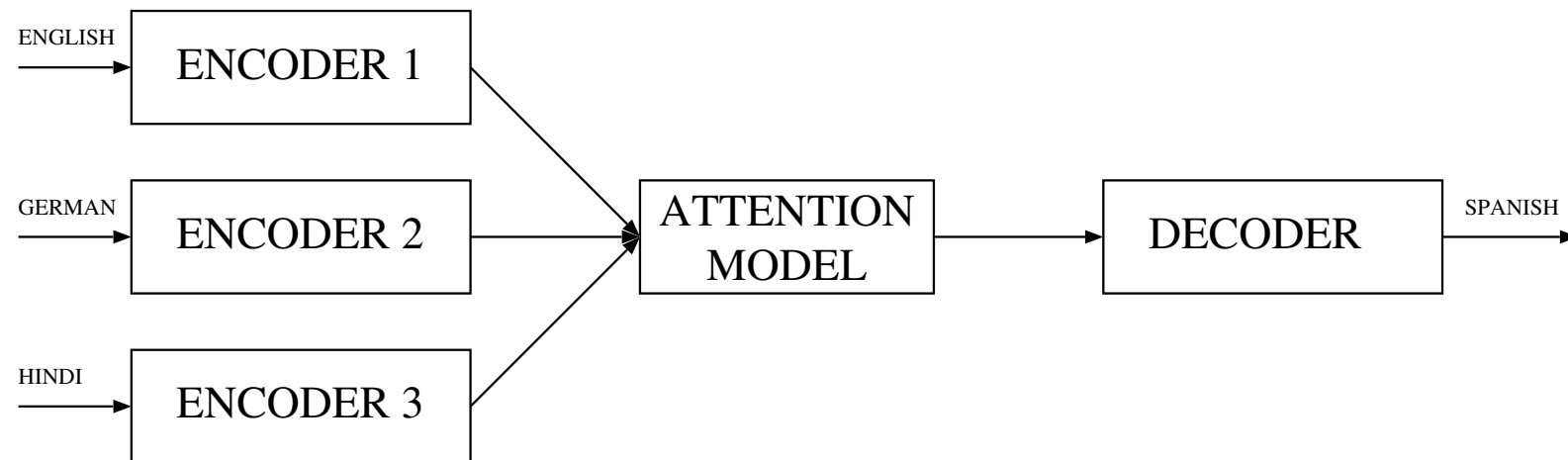
Scenarios for multilingual neural machine translation

[Dabre+ arXiv 2020]

- **Multi-way translation.** The goal is constructing a single NMT system for one-to-many, many-to-one or many-to-many translation using parallel corpora for more than one language pair. Parallel corpora are available for each language pairs.
- **Low resource translation.** (a) a high-resource language pair is available to assist a low-resource language pair. (b) no direct parallel corpus for the low-resource pair and a pivot language is used.
- **Multi-source translation.** Documents that have to be translated into more than one language [Zoph+ arXiv 2016].
- Multilingualism with RBMT and SMT: Interlingua approach.

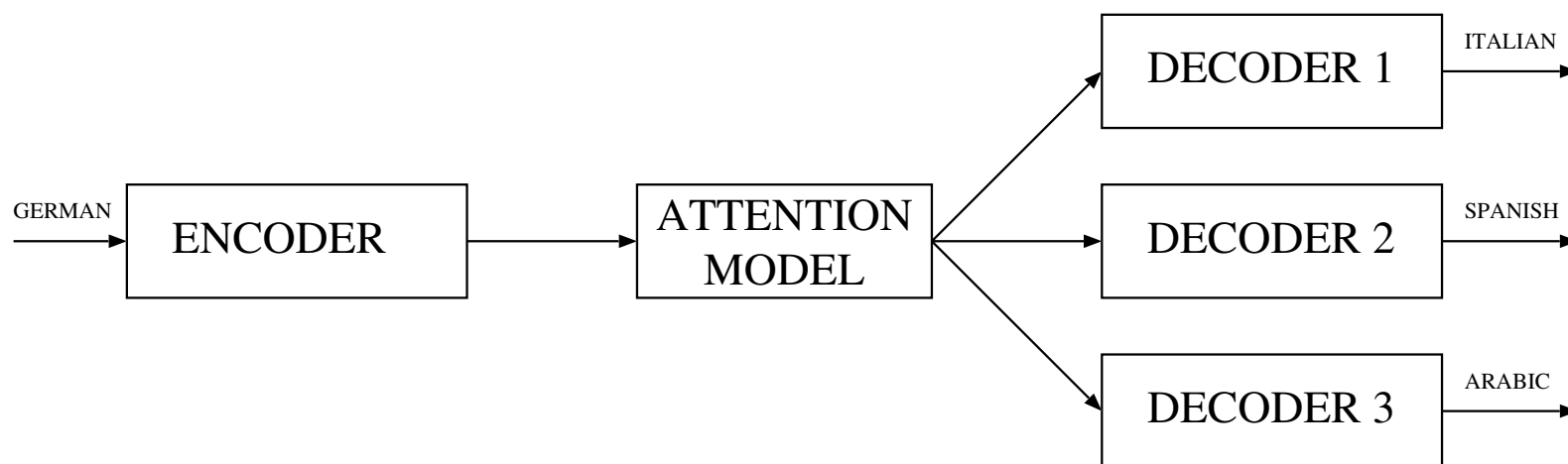
Multi-way neural machine translation (I)

Several languages to one language



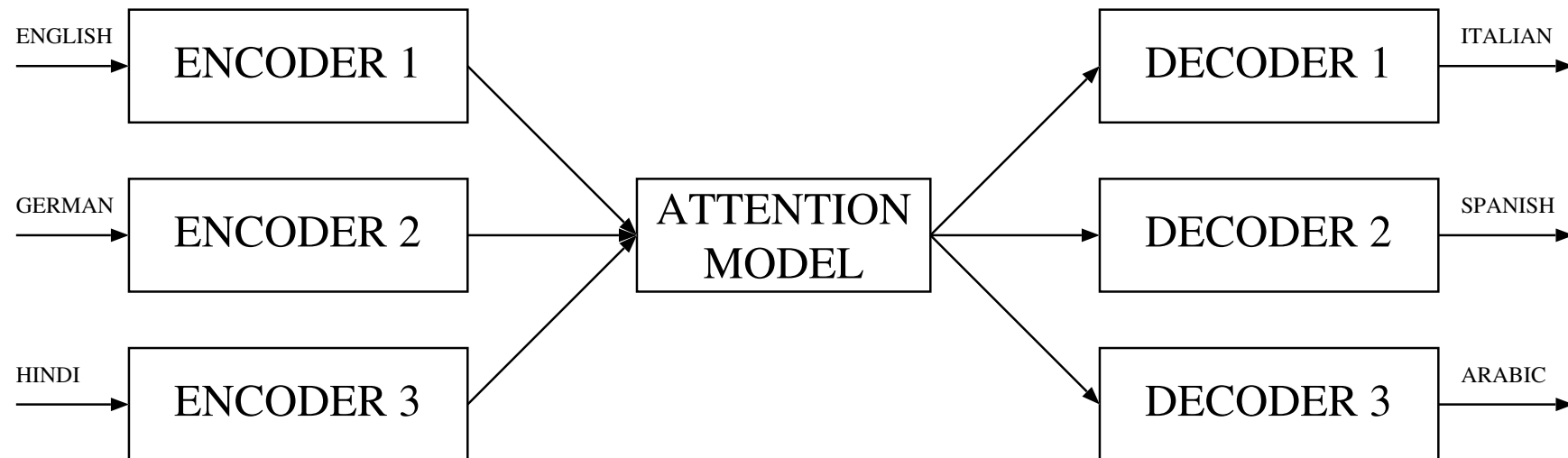
Multi-way neural machine translation (II)

One language to several languages



Multi-way neural machine translation (III)

Several languages to several languages

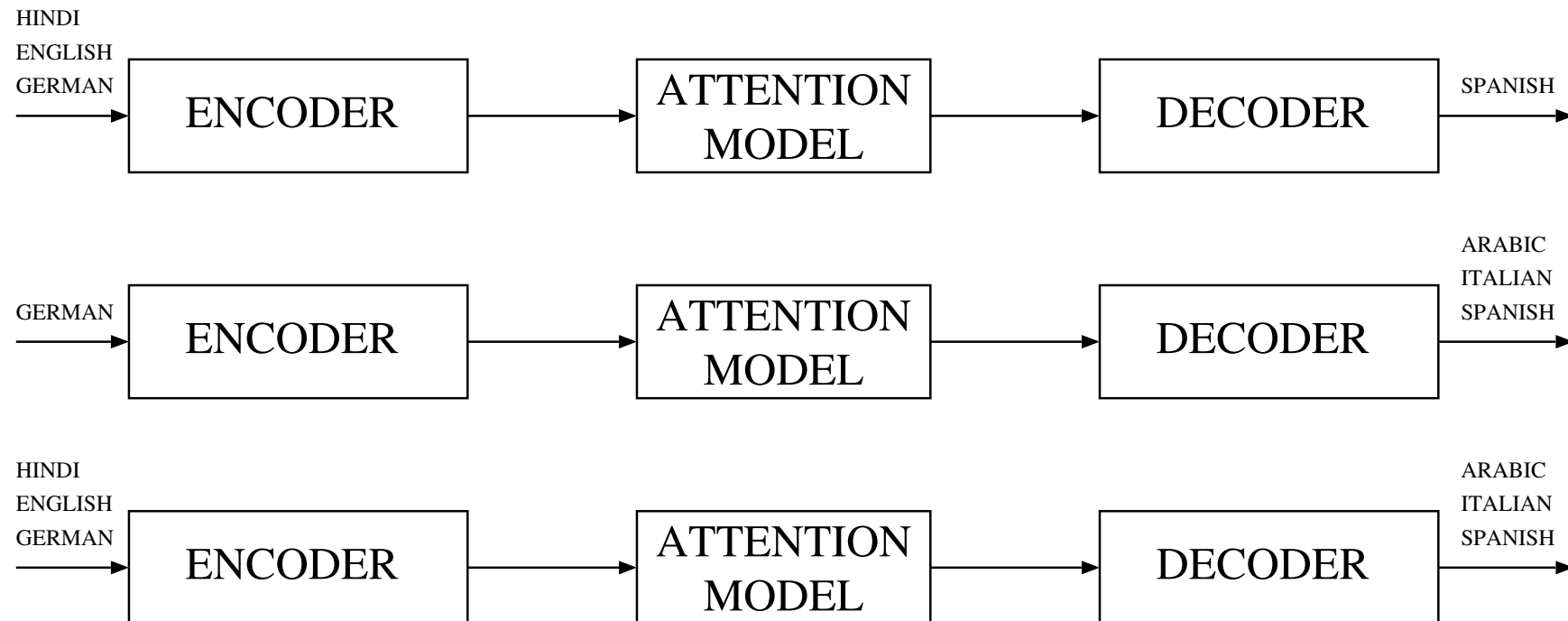


Models for multilingual neural machine translation

- One encoder for all source languages or one encoder for each source language.
- One decoder for all target languages or one decoder for each target language.

Multi-way neural machine translation

One encoder/decoder for all languages.



The language tag trick: A tag is added to each sentence to identify the language.

Training with MNMT [Dabre+ arXiv 2020]

- Regular training criterion for one language pair given a training bilingual corpus T :

$$\widehat{\mathbf{W}} = \operatorname{argmax}_{\mathbf{W}} \mathcal{F}_T(\mathbf{W}) \equiv \operatorname{argmax}_{\mathbf{W}} \sum_{(x_1^J, y_1^I) \in T} \log p(y_1^I \mid x_1^J; \mathbf{W})$$

- Training criterion for a set of language pairs ($L \subset S \times D$) pairs given a the corresponding training bilingual corpus T_l for $l \in L$:

$$\widehat{\mathbf{W}} = \operatorname{argmax}_{\mathbf{W}} \sum_{l \in L} \mathcal{F}_{T_l}(\mathbf{W}) \equiv \operatorname{argmax}_{\mathbf{W}} \sum_{l \in L} \sum_{(x_1^J, y_1^I) \in T_l} \log p(y_1^I \mid x_1^J; \mathbf{W})$$

- **Single stage parallel/joint training**
 - For models with separate encoders and decoders, each batch consists of sentence pairs for a specific language pair whereas for fully shared models, a single batch can contain sentence pairs from multiple language pairs.

MNMT for low-resources languages pairs [Dabre+ arXiv 2020]

- The high-resource and low-resource language pairs share the same target language. Jointly training both language pairs.
- Fine tune: First, they trained a parent model on a high-resource language pair. The child model is initialized with the parent's parameters wherever possible and trained on the small parallel corpus for the low-resource pair.
- Lexical Transfer.
- Syntactic Transfer

MNMT for unseen languages pairs [Dabre+ arXiv 2020]

- Zero-shot translation: The MNMT system has not been trained for the unseen language pair, but the system is able to generate reasonable target language translations for the source sentence.
- Zero-resource translation: Using a pivot language i.e. by synthetic corpus generation using a pivot language.

Some experimental results [Cuevas TFM 2020]

- From one language to several languages.
- One decoder for all target languages & one decoder for each target language.
- Toolkit: NMT-keras.
- Corpus: Europarl.
 - Source language: English.
 - Target languages: Spanish, German and French.

Some experimental results [Cuevas TFM 2020]

Experiment	Source language	Target language	BLEU	TER
Baseline	English	Spanish	23.6	58.0
		French	24.7	60.0
		German	15.0	68.3
One decoder	English	Spanish	21.3	61.0
		French	22.4	63.0
		Spanish	22.5	60.0
		German	14.4	73.6
		French	22.4	63.8
Two decoders	English	German	13.5	74.2
		Spanish	25.4	57.0
		French	25.5	59.0
		Spanish	25.4	56.9
		German	17.0	68.0
		French	26.6	59.0
		German	16.8	68.6

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Why document-based translation?

[Maruf+ arXiv 2019]

- In most of the applications, the goal is to translate a whole document and sometimes a paragraph.
- The common approach is to translate the document sentence by sentence as independent facts.
- However, sentence-based constraints do not deal with long-term dependencies as anaphora, ellipsis, word ambiguity, ...
- Therefore, the translation of a document sentence by sentence can suffer of a lack of coherence at document level.

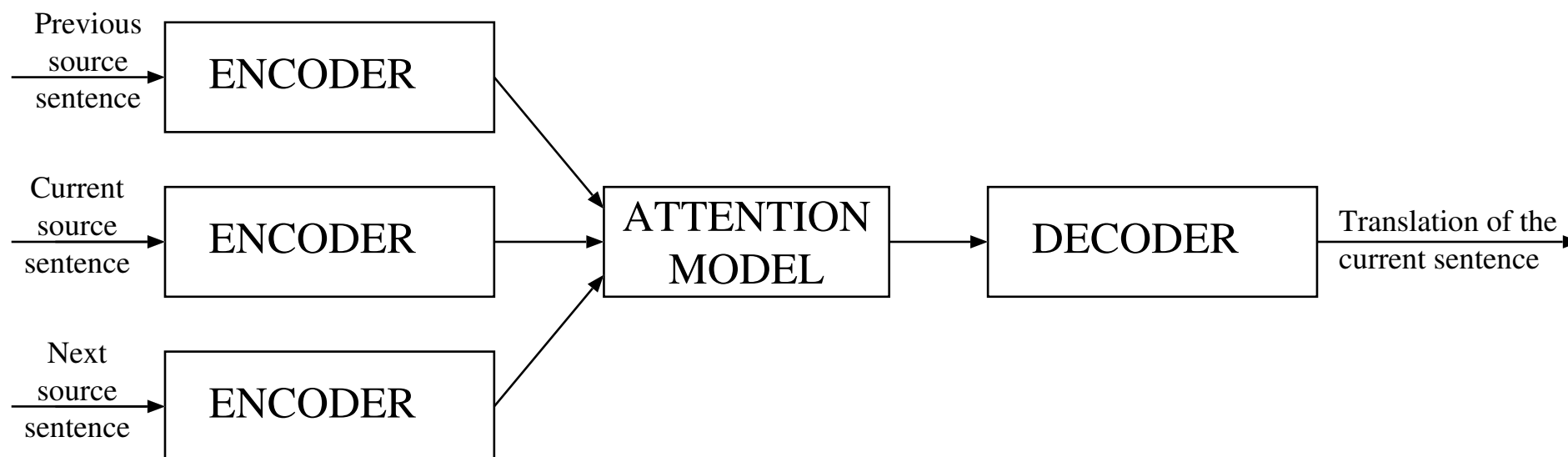
Approaches to document-based translation [Huo+ WMT 2020]

- SMT: Hard problem.
- NMT
 - Context without a modification of the architecture:
 - * Concatenate the current source and previous sentence.
 - Context via additional components:
 - * Additional context encoder and combining the representations from the current and the previous source sentence to feed the decoder.
 - * Additional context encoder and the current encoder feed to the cross-attention of the decoder in an independent way.
 - * Additional context encoder and the current encoder feed in parallel to two cross-attentions of the decoder and a combination of the output.
 - * A regular transformer for mapping source to target sentences plus a pretrained model BERT that deals with the concatenation of the current source and previous sentence.

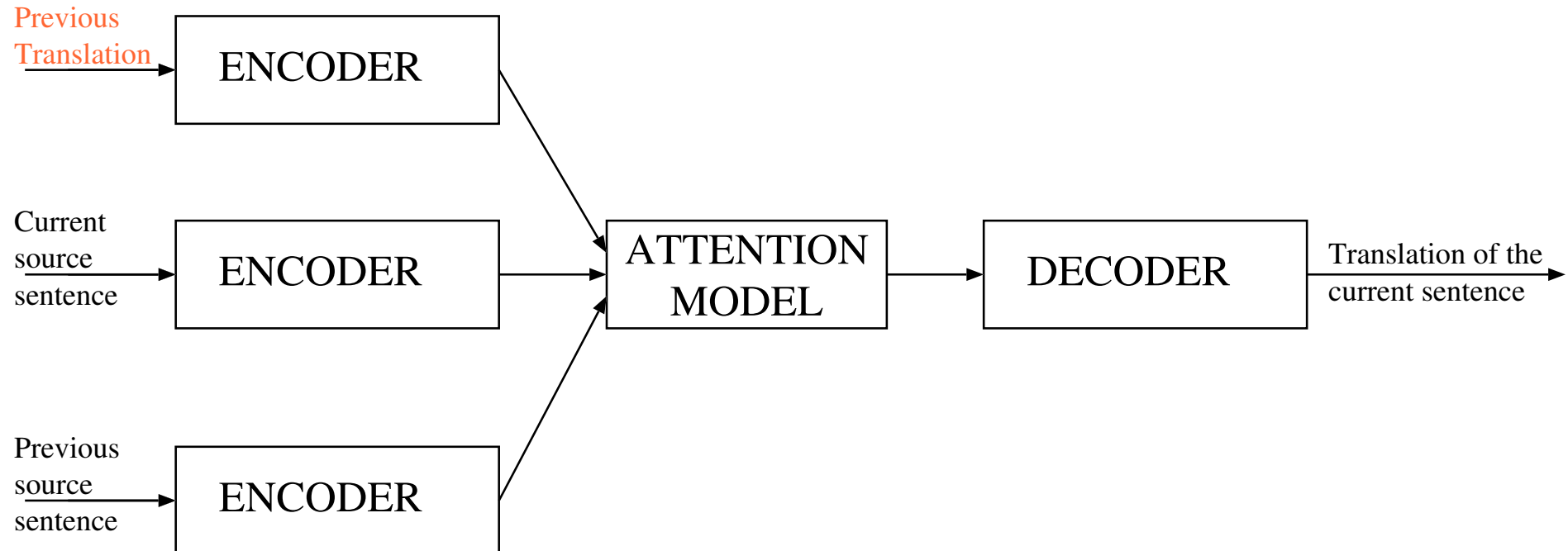
Approaches to document-based translation [Maruf+ arXiv 2019]

- SMT: Hard problem.
- NMT
 - Context without a modification of the architecture:
 - * Concatenate the current source and previous sentence.
 - Context via additional components:
 - * Additional context encoder and attention only.
 - * Additional context encoder and decoder.
 - Context via a two-pass approach:
 - * From context sentence encoder to document encoder.
 - Positive use of context during training.
 - * Discrimination to ignore irrelevant information.
 - * Using curriculum learning.

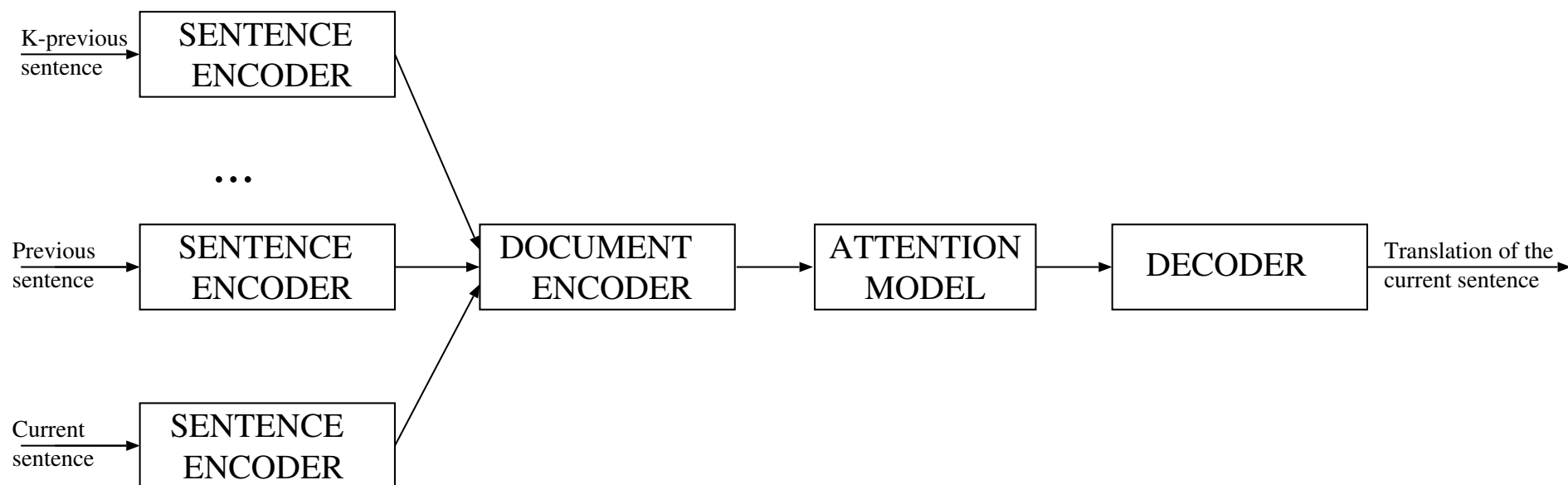
Additional components: source context in DNMT



Additional components: source & target context in DNMT



Additional components: two-level source context in DNMT



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Why monolingual corpora?

- Low resource languages:
 - Small bilingual corpora
 - Lack of bilingual corpora.
- Large availability of monolingual corpus in many languages

On the use of monolingual corpora for NMT [Gibadullin arXiv 2019]

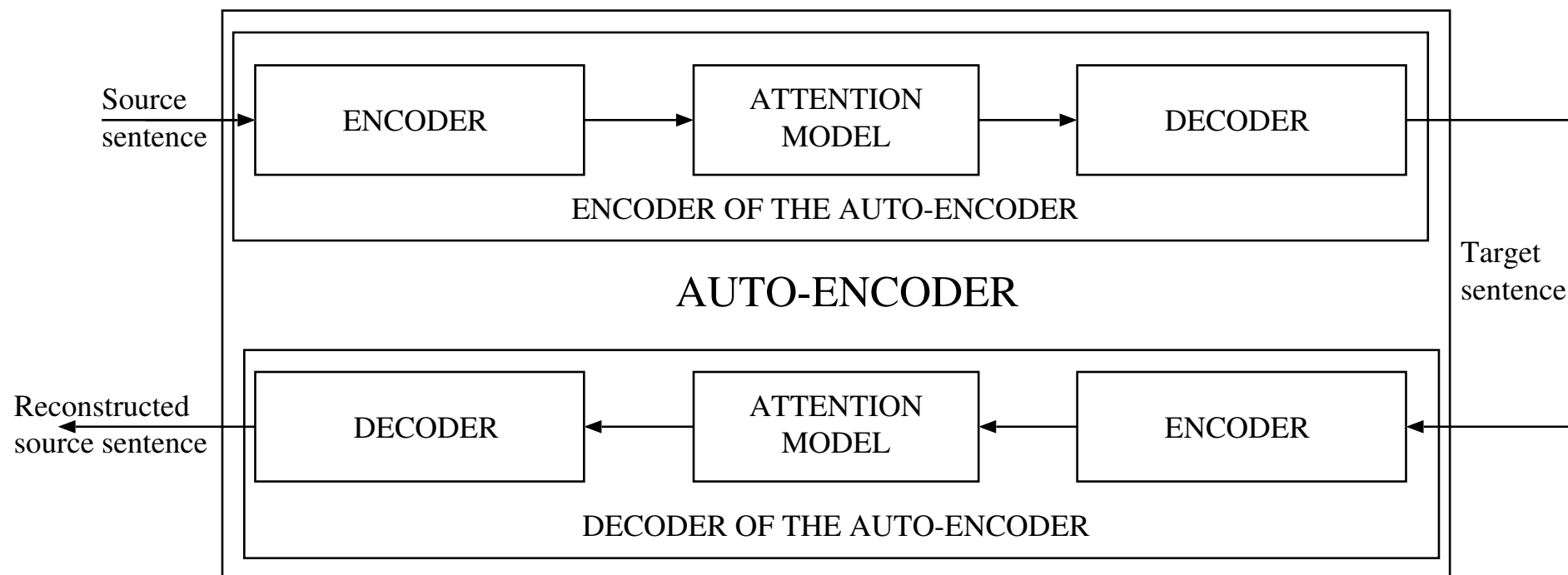
- Architecture independent methods:
 - Generate pseudo-parallel (synthetic) corpus using monolingual corpus.
 - Merge NMT models with a target language model from monolingual corpora.
- Architecture dependent methods:
 - Training with parameters freezing.
 - Integration of language modeling.
 - Pre-training with language models.
 - Multi-tasking.
- Fully unsupervised learning: From unsupervised multilingual word embeddings.

On the use of monolingual corpora for NMT [Gibadullin arXiv 2019]

Pseudo-parallel (synthetic) corpus

- Back-translation:
 - From a small parallel corpus, train a target to source model (T).
 - Translate the monolingual (target) corpus using T: Align source and target sentences (synthetic corpus)
 - True and synthetic corpora are merged to obtain a translation model.
- Round trip training:
 - Based on an auto-encoder.
 - Source-to-target translator is used as an encoder of auto-encoder and target-to-source as a decoder of the auto-encoder.
 - The whole training objective of the method is to maximize the likelihoods of source-to-target and target-to-source models on parallel corpus, and reconstruction likelihoods of auto-encoders on monolingual corpora.

Round Trip Training



On the use of monolingual corpora for NMT [Gibadullin arXiv 2019]

Merge with a separate language model

- Using monolingual target model to train a language model.
- Merge the target language model with translation models in the inference.
 - Shallow fusion: for each predicted target word, sum the probability of the translation model and the language model. Translation models and language models are trained in a separate way.

Architecture dependent methods [Gibadullin arXiv 2019]

- Training with parameters freezing:
 - Forward-translation: Pseudo-parallel corpus from monolingual source data and freezing decoder parameters when pseudo-parallel data is used.
 - Dummy input: Each target monolingual sentence is associated to a single-word null to produce the pseudo-parallel corpus. The parameters of the encoder and attention model is freezing when that pseudo-parallel corpus is used.
- Integration of language modeling in the training process:
 - Deep fusion: The hidden state of the LM and the hidden states of the decoder are concatenated.

Architecture dependent methods [Gibadullin arXiv 2019]

- Pre-training with Language Models: Pre-training of the RNN model: A pre-trained source LM is used as the encoder and a pre-trained target LM is used as the decoder. Parallel corpus is used for fine-tuning.
- Multi-tasking: Two decoders are used, one for translation and the second one for reordering the source sentence, The reordered source sentences for reordering are obtained by the application of some rules.

Fully unsupervised learning [Gibadullin arXiv 2019]

- Bilingual dictionary induction, and the pseudo-parallel corpus is obtained by applying the bilingual dictionary to the monolingual corpus.
 - Linear transformation of source word embeddings to target word embeddings. In this common space, translations of words in one language can be found by searching nearest neighbors among the words from another language. The transformation matrix can be found using some small seed dictionary or even without it [Artetxe ACL 2018].

Some experimental results [Castellanos TFM 2020]

- Using back-translation to generate pseudo-bilingual corpus.
- Merge bilingual corpus and pseudo-bilingual corpus
- Toolkit: openNMT.
- Corpus: Europarl.
 - Source language: English.
 - Target languages: Spanish.

Some experimental results [Castellanos TFM 2020]

Experiment	Corpus size			BLEU
	Bilingual	Pseudo-parallel	Total	
Baseline	200K		200K	13.4
	400K		400K	16.8
	600K		600K	18.8
	800K		800K	20.1
	1,000K		1,000K	20.6
	2,000K		2,000K	22.3
+ back-translations	200K	200K	400K	17.3
	400K	400K	800K	19.1
	600K	600K	1200K	20.4
	800K	800K	1600K	20.7
	1,000K	1,000K	2,000K	21.2

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Why linguistic knowledge?

- YES?
 - There are many linguistic knowledge available.
 - The bilingual training data can be better exploited.
- NOT?
 - Many linguistic knowledge is hard to formalize.
 - The generation of new linguistic knowledge requires great human effort.

Syntax in NMT [Zhang NAACL 2019]

- Syntactic trees could offer long-distance relations in sentences
- Formulation:

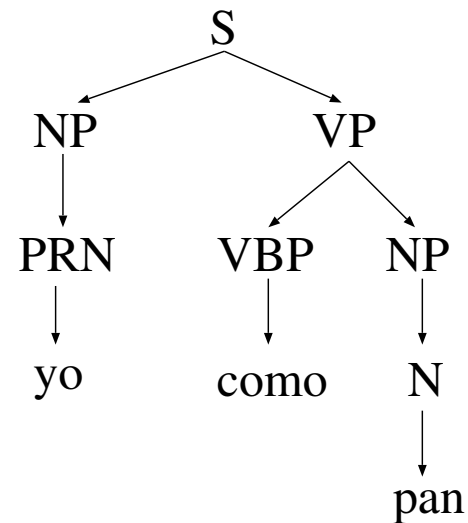
$$\begin{aligned}\hat{y}_1^I &= \operatorname{argmax}_{I, y_1^I} p(y_1^I \mid x_1^J, tr(x_1^J)) \\ &= \operatorname{argmax}_{I, y_1^I} \prod_{i=1}^I p(y_i \mid y_1^{i-1}, x_1^J, tr(x_1^J))\end{aligned}$$

- Approaches:
 - Tree-structured recurrent neural network (Tree-RNN) [Yang EMNLP 2017]
 - Tree-Linearization: traverse a constituent tree [Li ACL 2017].

Tree-Linearization [Li ACL 2017]

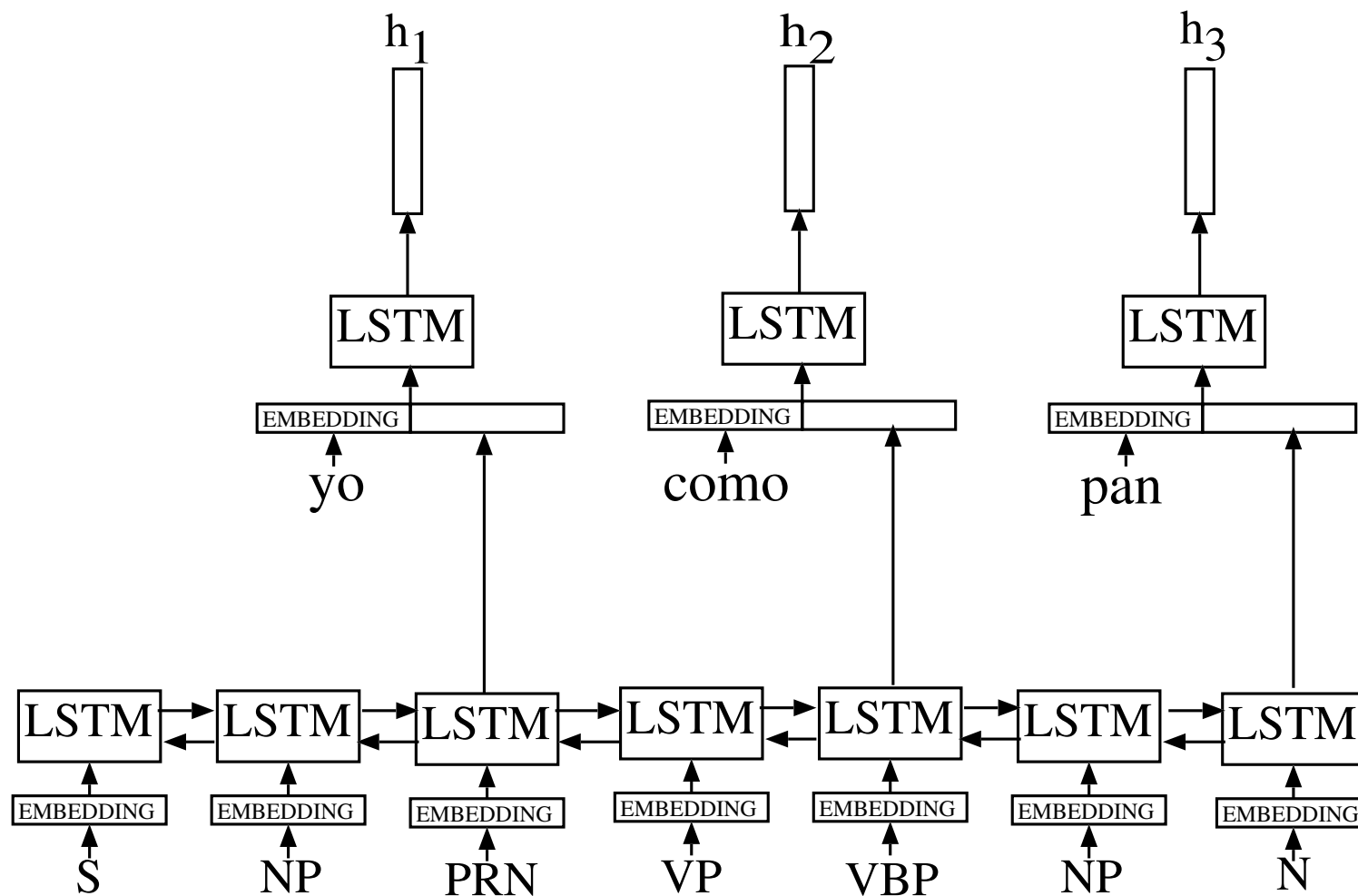
- Sentence: “yo como pan”

- Syntax:

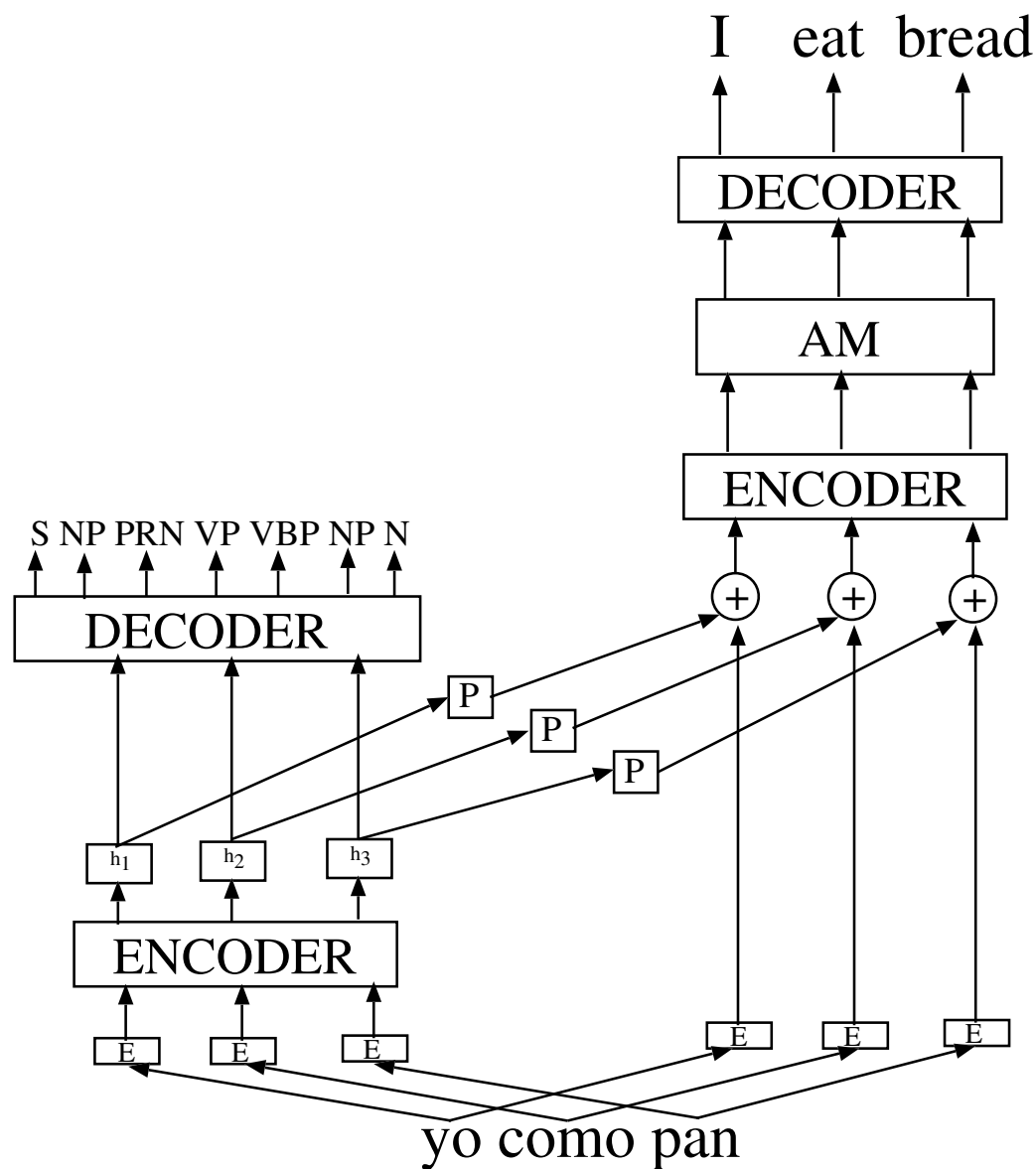


- Linearization: “S NP PRN VP VBP NP N”

Tree-Linearization: Hierarchical encoder [Li ACL 2017]



Syntax-Aware word representations [Zhang ACL 2019]



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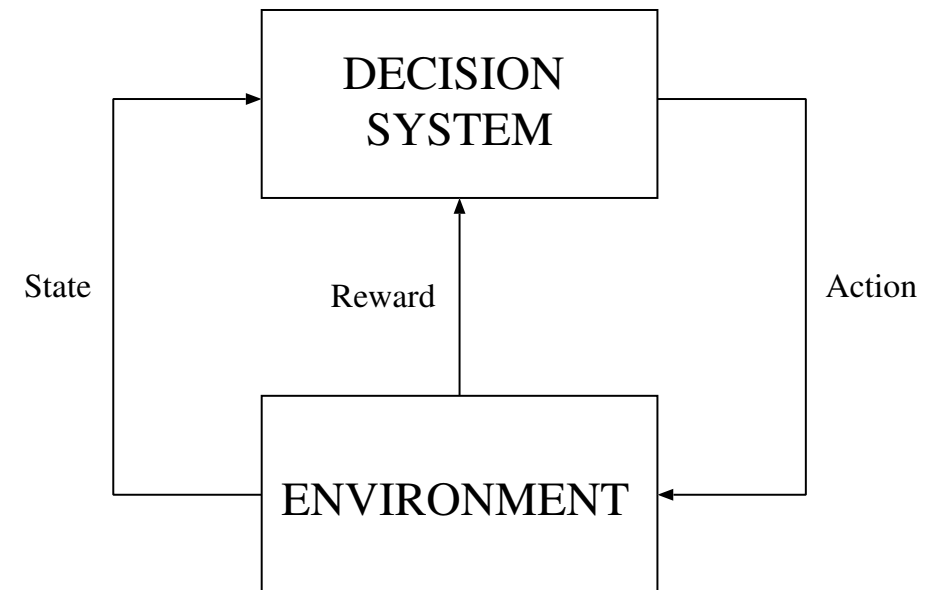
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Why reinforcement learning in NMT?

- Fine-tuning for optimizing metrics as BLEU, TER, ...
- A new framework for interactive machine translation

Markov decision process

- Set of **states** $\mathcal{Q}$.
- Set of **actions** $\mathcal{A}$.
- **Transition probability** (*Markovian*):
 $\tau(q' \mid q, a), \quad q, q' \in \mathcal{Q}, \quad a \in \mathcal{A}.$
- **Reward function** $r : \mathcal{Q} \times \mathcal{A} \times \mathcal{Q} \rightarrow \mathbb{R}.$



Markov decision process

- In step t the system is in state $q \in \mathcal{Q}$. The agent chooses one action $a \in \mathcal{A}$ and a new state $q' \in \mathcal{Q}$ is achieved (according to $\tau(Q_{t+1} = q' \mid Q_t = q, A_t = a)$), with a reward $r(q, a, q')$.
- An **episode** h is a sequence of steps (with “finite horizon” or “episodic”):

$$h = [q_1, a_1, q_2, a_2, \dots, a_T, q_{T+1}]$$

An **episode** with “infinite horizon” or “continuous” is $h = [q_1, a_1, q_2, a_2, \dots]$

- The **gain** or accumulated reward for the episode h is:

$$G(h) = \sum_{t=1}^T \gamma^{t-1} r(q_t, a_t, q_{t+1}) = \sum_{t=1}^T \gamma^{t-1} r_t$$

where γ ($0 \leq \gamma < 1$) is **discount factor**.

Policy

- A **policy** is an agent strategy to choose the actions of the successive steps. It can be defined as a probabilistic distribution $\pi(A_t = a \mid Q_t = q)$ (stochastic policy) or as a function $(\pi : \mathcal{Q} \rightarrow \mathcal{A})$ (deterministic policy). ($\pi(a \mid q) = \delta(a, \pi(q))$)
- The probability of a episode h , given a policy π is

$$p_{\pi}(h) = p_i(q_1) \prod_{t=1}^T \tau(q_{t+1} \mid q_t, a_t) \pi(a_t \mid q_t) = p_i(q_1) p_{\pi}(h \mid Q_1 = q_1)$$

where $p_i(q_1)$ is the probability of the episode h starts in the state q_1 .

- An **optimal policy** π^* is:

$$\pi^* = \operatorname{argmax}_{\pi} \mathbb{E}_{p_{\pi}(h)} [G(h)] = \operatorname{argmax}_{\pi} \sum_h p_{\pi}(h) G(h)$$

Policy

- Given a model (τ, r) estimated from a set of episodes, compute the optimal policy π^* :
 - **Policy Iteration** algorithms (based on states or states-action values)
- If the model (τ, r) is unknown, compute the optimal policy π^* directly from the episodes (**model free**):
 - **Temporal Difference** Learning (for computing state and state-action values)
 - **Policy gradient** $\pi(a \mid q; \theta)$: REINFORCE or ACTOR-CRITIC.
 - **Least-Squares Policy Iteration**.
 - **Deep Q-network**: a deep network for computing state-action value.

Reinforcement learning in NMT [Wu EMNLP 2018]

- Training NMT with Reinforcement Learning: directly optimizing the evaluation measure at training time.
 - State: (y_1^{i-1}, x_1^J)
 - Policy: $p(y_i | y_1^{i-1}, x_1^J)$
 - Action: choose the next translation word y_i
 - Reward: the BLEU at the end of translation $BLEU(y_1^I, \bar{y}_1^I)$ where $\bar{y}_1^I$ is the reference translation.
 - * Problem: reward at the end of translation.
 - * Solution: $r_i(y_i, \bar{y}_1^I) = BLEU(y_1^i, \bar{y}_1^I) - BLEU(y_1^{i-1}, \bar{y}_1^I)$ (Reward shaping)
 - Combine MLE and RL objectives
- Training NMT with Reinforcement Learning and monolingual data

Reinforcement learning for INMT

[Lam EAMT 2018] [Lam MTSUMMIT 2019]

- The user can “DELETE”, “SUBSTITUTE” or “KEEP” a translated word:
 - State: (y_1^{i-1}, x_1^J)
 - Policy: $p(y_i \mid y_1^{i-1}, x_1^J)$
 - Action: choose the next translation word y_i
 - Reward: $r_i(y_i) = \begin{cases} 0.5 & \text{if SUBSTITUTE/KEEP} \\ -0.1 & \text{if DELETE} \end{cases}$
- Learning algorithm for the policy: ACTOR-CRITIC.
- The policy is updated from edits if the entropy (uncertainty) of the action is high enough.

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More ...

- Computational aspects: Efficiency, parallelism, multiple GPUs.
- Reduce the model size [Banik+ IEEE Access 2018].
- More explanatory models.
- Linguistic knowledge from continuous representations [Manning+ PNAS 2020].
- Data selection.
- More learning algorithms: incremental learning,
- Automatic/human evaluation.
- Confidence measures.
- Ensemble decoding.
- The use of specific glossaries (place holders)
- Machine translation for machines [Tebbifakhr+ EMNLP 2019].
- Knowledge distillation.
- ...

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